1 Blockchain basics

This chapter covers

- Understanding blockchain
- Discovering decentralized system infrastructure
- Exploring distributed ledger technology
- Analyzing trust-enabling protocol
- Motivating blockchain applications with real-world scenarios

In the latter part of 2008 and early 2009, centralized systems of the world's financial markets--enabled by large intermediaries such as banks and investment firms--failed and began to crumble. Trust in these systems eroded, and panic set in all over the world with the collapse of financial markets. It was at this juncture that a mysterious person or persons introduced to the world a working model of a peer-to-peer decentralized digital currency system (with no central authority or administration) called Bitcoin. The trust intermediation in this system was realized via software that would later be named blockchain. Blockchain provided the software-based verification, validation, recording, and integrity essentials for currency transfers.

Even though Bitcoin appeared to have launched suddenly in 2009, the idea of a working digital currency has been a quest since the dawn of computing. Bitcoin's blockchain